NumPy Mini-Project: House Price Prediction

1. Introduction

The objective of this project was to build a simple house price prediction model using Python and NumPy. The model predicts the price of a house based on its size in square feet.

2. Dataset

The dataset contains 11 houses, with two features:

- **Input (X):** House size in square feet.
- **Target (y):** House price in lakh rupees.

The dataset was created manually for this project. The first 8 houses were used for training, and the last 2 houses were used for testing.

3. Data Analysis and Preparation

NumPy was used to create and manipulate arrays, calculate statistical measures, and split the data into training and testing sets.

The following operations were performed:

- Calculated the mean, median, minimum, maximum, and standard deviation.
- Split the dataset into training and testing sets.
- Shuffled the dataset using random indices while preserving the correct size–price pairs.

4. Linear Regression Model

A simple linear regression model was developed using the equation:

Predicted Price = w × House Size + b

Where:

- w is the slope, representing the price change per square foot.
- b is the intercept.

The slope and intercept were calculated using the training data. The resulting model was approximately:

Predicted Price = 0.03 × House Size

The intercept was approximately zero.

5. Predictions and Evaluation

The model was evaluated using two test houses.

House size	Actual price	Predicted price
2800 sq. ft.	₹84 lakh	₹84 lakh
3000 sq. ft.	₹90 lakh	₹90 lakh

The model's predictions matched the actual prices in this dataset.

Two evaluation metrics were calculated:

- **Mean Absolute Error (MAE):** Approximately 0 lakh.
- **Mean Squared Error (MSE):** Approximately 0 (in squared lakh units).

These results indicate that the predictions were accurate for the selected test houses.

6. Baseline Comparison

A baseline model was created that always predicts the average price of the training houses, without considering their sizes.

- Baseline average price: ₹41.25 lakh.
- Baseline MAE: ₹45.75 lakh.
- Linear regression MAE: Approximately 0 lakh.

The linear regression model performed better than the baseline model on the test data.

7. Challenges and Learning Outcomes

The project helped me understand:

1. NumPy arrays, indexing, slicing, and array operations.
2. Statistical calculations using NumPy.
3. Training and testing data.
4. The basic working of linear regression.
5. Predictions and error metrics such as MAE and MSE.

6. Baseline models and their importance in evaluating machine learning models.
7. Shuffling data while maintaining the correct relationship between inputs and targets.
8. Handling invalid user input using `try-except`.

8. Limitations

The dataset is small and follows a perfectly linear relationship between house size and price. Real-world house prices also depend on factors such as location, age, number of rooms, and condition. Therefore, the model's results do not demonstrate how accurately it would predict prices in a real housing market.

9. Conclusion

This project provided practical experience in using NumPy to implement a simple linear regression model from scratch. I learned how to prepare data, calculate model parameters, make predictions, evaluate errors, compare the model with a baseline, and shuffle data correctly. The project strengthened my understanding of the basic concepts behind machine learning.